

UNIVERSITY OF TORONTO

FACULTY OF APPLIED SCIENCE AND ENGINEERING

Short Answer

November 2nd 2021

First Year

APS110 ENGINEERING CHEMISTRY AND MATERIALS SCIENCE

Exam Type: B (Closed Book, single aid sheet provided)

Examiners: F Gu, C Jia, S Ramsay

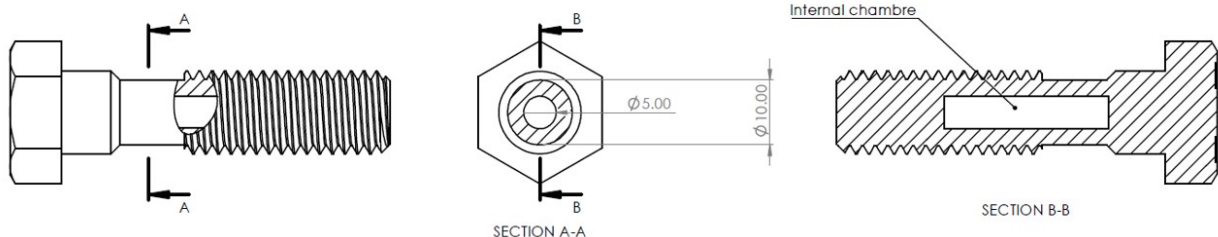
All numerical responses must be expressed in the most appropriate units, including appropriate prefixes (ex. GPa rather than 10^9 Pa), and be written with appropriate significant figures.

NAME: _____

STUDENT NUMBER: _____

Question	Points	Score
1	10	
2	8	
3	8	
Total:	26	

1. A pyrotechnic explosive bolt is a type of mechanical fastener that can be used to secure a component that must be quickly released at a later time. This type of bolt is commonly used to secure rocket stages, modules of spacecraft, or hatches that must be released and separated quickly. Assume that the bolt we are considering is made from a steel alloy having a yield strength of 350 MPa and an ultimate tensile strength of 420 MPa and has the critical dimensions given in the figure below, in mm.



(a) (3 points) What is the maximum load that this bolt can support without deforming permanently?

(b) (3 points) What is the minimum load that will be required to fracture this bolt?

(c) (4 points) The explosive is placed within the internal chambre and expands upon detonation, applying a pressure to the inner walls of the internal chambre. Assuming this pressure only acts to produce a tensile force against the circular ends of the internal chambre, what is the minimum pressure, in MPa, that must be applied by the explosive?

2. (8 points) Some sources believe that, on his recent trip to space, William Shatner made an unscheduled spacewalk to collect some samples of several mysterious substances seen sticking to the outside of his New Shepard spacecraft. Upon his return from space he analysed the samples and tabulated all of the information he could obtain, shown in the table below. Using the data in the table, calculate the missing data for each mysterious substance. Assume each sample is a single element, with the exception of Chartreuse Matter, which is an ionic compound. Show your work.

Name	Radius (pm)	Crystal Structure	Molar Mass (g/mol)	Density (g/cm ³)
Pink Matter		FCC	81.4	8.96
Puce Matter	152	Simple Cubic		8.41
Periwinkle Matter	210	BCC	52.8	
Chartreuse Matter	$R_A = 152$ $R_C = :$	Rock Salt	$A_A = 101$ $A_C = 110$	4.98

3. (8 points) In the space below, describe how the strength of a metal can be increased by using each of the following crystalline imperfections: alloying elements, dislocations, grain boundaries, second phase particles.

This is the end of the written portion of the assessment.